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Generative Artificial Intelligence in the Legislative Process: Perspectives of Knowledge Generation

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Abstract

Despite the growing interest in the application of AI in the public sector, there is a lack of in-depth studies on how generative AI can be used specifically to improve the legislative process. This article aims to fill this gap by investigating the potential of tools as an example to automate routine tasks, thus contributing to the advancement of knowledge about the implications of generative AI in the drafting and analysis of laws. Among the proposed uses are the automatic generation of drafts, summaries, and legislative opinions, which can speed up and facilitate the work of advisors and parliamentarians. Furthermore,

these AIs can more easily identify potential inconsistencies or incoherences in law proposals, resulting in clearer and more objective texts. The article also highlights the need to establish ethical guidelines for the responsible use of technology in the legislative branch. The research was conducted using the integrative literature review method, and articles collected from the Ebsco, Scopus, and Web of Science databases were analyzed. Within this context, the research aims to provide new insights for the debate on how parliaments can incorporate these new technologies in the best possible way into their processes.

Keywords – artificial intelligence, legislative process, knowledge generation, public sector

Paper type – Academic Research Paper

1 Introduction

It is undeniable that generative artificial intelligences are at the forefront of current technological progress. This advancement discusses not only the potential increase in productivity but also the inherent challenges, such as the need for organizational adaptations and regulations, which are essential to fully leverage the potential of generative AI technologies (Veloso, 2023).

Generative AIs bring efficiency to processes such as the foundation of proposals and regulatory impact analysis (Bassante, 2023). Considering their transformative potential, it is relevant to explore applications in various areas, including the Legislative Power.

In this context, generative AI can automate the production of legislative documents such as drafts, opinions, and reports, increasing the agility of the work of advisors and parliamentarians. These tools can also act as virtual assistants, performing repetitive tasks and freeing up more time for employees to engage in strategic functions (Bassante, 2023).

AI offers various practical applications in the public sector, improving efficiency and communication. For example, dialogue systems, such as chatbots, are used to answer citizens' questions about public services, reducing waiting times (Janssen et al., 2020).

The analysis of large volumes of open data by AI helps identify useful patterns in policy formulation in areas such as health and education (Susha et al., 2020). Furthermore, AI can create illustrations and explanatory videos to convey government information more clearly and attractively (McGrenere et al., 2020).

Indeed, the identification of patterns in legal texts by generative AI, as discussed by Alencar (2023), reveals its potential to increase consistency in legislative proposals, minimizing ambiguities and contradictions. Moreover, this

technology has the capability to improve communication between citizens and authorities, contributing to the strengthening of participatory democracy.

Given the above, the present article's research problem is to assess how generative AIs can increase the efficiency and quality of legislative work.

Despite the growing interest in the application of AI in the public sector, there is a lack of in-depth studies on how generative AI can be used specifically to improve the legislative process. This article aims to fill this gap by investigating the potential of tools as an example to automate routine tasks, thus contributing to the advancement of knowledge about the implications of generative AI in the drafting and analysis of laws.

Regarding specific objectives, it proposes to: explore how generative AIs can be applied to automate the production of legislative documents and facilitate the analysis of large volumes of information; identify possible ethical and regulatory challenges in implementing these technologies in the legislative context; compare international initiatives of AI application in the legislative process, highlighting lessons learned and best practices.

While the introduction of AI in the Legislature can improve speed and accuracy, it is necessary to evaluate how to ensure that the benefits outweigh any potential problems, always aiming for the public interest.

Authors such as Xavier (2023), Badaró and Campos (2023), Lucke et al. (2023), Salgado-Criado and Fernández-Aller (2021), and Pilving (2023) bring important perspectives on these issues, such as: reliability, bias risks, the need for human supervision, protection of users' data, and ethical and legal implications.

Also, it is interesting to highlight how AIs can be used responsibly and efficiently, emphasizing the importance of measures like data anonymization, clear indication of AI use in legal documents, and the creation of policies to guide the use of such technologies.

Finally, it is interesting to note that, in the advent of the digital era, generative artificial intelligence (AI) emerges as a promising tool for transforming various sectors, including the Legislative Power.

Indeed, although there are extensive discussions on the potential benefits of AI in automating and optimizing processes in corporate and industrial environments, a significant gap remains in understanding how these technologies can be specifically applied to improve the efficiency, quality, and transparency of legislative work.

Thus, besides the introduction, the article presents five sections. Section 2 addresses two theoretical frameworks on: 1) Integration of Artificial Intelligence and Knowledge Management in the Public Sector; and 2) Generative Artificial Intelligence in the Legislative Branch. The third section describes the research's methodological aspects. The fourth section brings the analysis and discussion of

the results. Finally, the last section presents the final considerations, limitations, and recommendations for future work.

2 Theoretical Framework

2.1 Integration of Artificial Intelligence and Knowledge Management in the Public Sector

Knowledge management has proven to be a critical factor for the success of digital transformation in public organizations. Thus, three benefits of knowledge management in digital transformation are identified: a) the enhancement of governmental competency; b) the improvement in the quality of governmental services; and c) the promotion of healthy governmental development (Alvarenga et al., 2020).

Therefore, the use of technology combined with the systematic use of knowledge increases and provides leverage and articulation of employees' skills and knowledge with the support of information technology.

On another vertex, a knowledge-intensive organization can be considered as an organization whose operations depend on specialized knowledge (Mazorodze and Buckley, 2020).

In turn, knowledge-intensive organizations are characterized by the use of highly skilled labor, as well as intensive use of technology (Khan et al., 2020).

In the context of knowledge-intensive organizations, digitalization and automation are strongly characterized. Mazorodze and Buckley (2020, p. 1) state that "a knowledge-intensive organization is an organization whose operations depend on specialized knowledge" (text translated by the authors). Khan et al., 2020, affirm that knowledge-intensive organizations are characterized by the use of highly skilled labor, as well as intensive use of technology.

As a result, Knowledge Management also saw a shift in how knowledge is captured, stored, and shared, with an increasing emphasis on digital solutions. Thus, modern Knowledge Management involves converting these large volumes of data into actionable insights and valuable knowledge (Santos, 2020).

With this, Artificial Intelligence (AI) and machine learning allow organizations to analyze patterns, predict trends, and even make data-based decisions.

Therefore, it is evident that AI can be used to improve the efficiency of Knowledge Management processes, automating routine tasks and providing deeper insights (Santos, 2020).

Alvarenga et al. (2020) discuss that knowledge management plays a critical role in the success of digital transformation in public organizations, bringing benefits

such as competency enhancement, improvement in the quality of services provided, and promotion of healthy organizational development.

With this focus, the Legislative Branch is also undergoing an increasing incorporation of digital technologies, needing to deal with large volumes of information, making it relevant that this modernization be accompanied by strategic knowledge management.

In 2023, Artificial Intelligence (AI) continues to evolve and transform various sectors, with significant advances in learning, decision-making, and task automation capabilities. According to Awari, AI is expected to drive areas such as customer service, data analysis, autonomous vehicle operation, health and medicine, and smart agriculture.

Moreover, AI will play a crucial role in optimizing processes in manufacturing and the financial sector, among others (Awari, 2023).

Similarly, McKinsey assesses that 2023 is a year of explosive growth for Generative AI (GenAI), with high expectations for significant changes in various sectors. These changes are particularly expected in the areas of technology and financial services. Additionally, addressing the risks associated with the use of GenAI is a challenge for many organizations (McKinsey & Company, 2023).

Thus, public sector organizations are increasingly interested in using data science and artificial intelligence capabilities to implement policies and generate efficiencies in environments of high uncertainty (Slava et al., 2018). Indeed, the adoption of new mechanisms, notably the use of artificial intelligence in the context of Big Data (which has the potential to exponentially increase the level of efficiency in public service delivery), may entail, for example, the review of the number of public agents that the State needs to fulfill certain tasks (Araujo, et al., 2020).

For this to occur, machine learning enables the recognition of patterns and deviations in human behavior. And thus, it is capable of making diagnoses, prognoses, and taking (or underpinning) decisions on complex issues. These are pieces of information that, if well processed, can represent a competitive differential for private sector companies, as well as an optimization in the field of public management (Araujo, et al., 2020).

Indeed, the tool is trained based on a vast dataset from web pages and, through inferences and probabilities, can answer questions, translate and summarize texts, generate codes, create poems, among others, in a manner very similar to human productions, constituting the group of so-called generative artificial intelligences (IAs) that can create new data based on patterns and structures learned from existing data (Badaró, et al., 2023).

Bassante (2023) identifies three main areas in which generative AI can significantly benefit public institutions: first, the transformation of the workforce,

preparing and training public employees for the AI era, including the adoption of technologies such as natural language processing to increase productivity; second, citizen support, using AI to improve public services and facilitate bureaucratic processes; and third, digital transformation, highlighting the crucial role of AI in the digitalization of the public sector.

Governments face challenges in modernizing legacy systems, but AI could be a solution, especially when integrated with cloud technologies. This can optimize data storage, improve pattern detection, automate computer programming, and strengthen cybersecurity, reducing the skills gap and exploring emerging talents (Bassante 2023). However, it is necessary to discuss the impact of AI on society, addressing the ethical challenges associated with the use of AI, such as algorithmic bias and privacy (Cortés et al., 2021).

Xavier (2023) highlights several positive applications of generative AI in the public sector, including: improving public digital services through interfaces such as chatbots, which facilitate access to information and increase citizen satisfaction; enhancing public efficiency by assisting employees in repetitive and creative tasks, and directing resources towards more strategic actions; promoting creativity and innovation by using AI to generate ideas in sectors such as urban planning and health, encouraging social collaboration; and, finally, strengthening transparency and citizen participation by making public data more accessible.

However, the mentioned author also warns about the associated risks, such as the creation of fake content and the manipulation of information, which can affect the credibility of the public sector. He emphasizes the importance of using AI in an ethical and responsible manner, suggesting measures such as keeping up with technology trends, training employees, defining regulatory policies, establishing partnerships, and involving citizens in discussions about AI (Xavier 2023).

It is also crucial to address the legal and ethical issues related to the use of Artificial Intelligence, especially in contexts such as advertising and copyright (Corrêa, 2023).

2.2 Generative Artificial Intelligence in the Legislative Branch

Undoubtedly, the use of AI in public administration will bring quality and efficiency to service delivery. Among the three branches of government, the Executive has taken the lead, already applying this type of technology in various segments, such as customer service, data analysis, autonomous vehicle operation, health and medicine, and smart agriculture (Awari, 2023).

In the Judiciary, the topic of AI is increasingly on the agenda, especially after the implementation of case management platforms in various courts around the

country. In August 2018, the Supreme Federal Court (STF) began using an ambitious project developed in partnership with the University of Brasília (UNB), called "Victor", which aims to employ artificial intelligence to improve and expedite the judicial analysis of cases in the STF (Brasil, 2018).

The adoption of artificial intelligence (AI) in the Legislative Branch, although in its initial stages, already shows notable progress, such as the "Ulysses" system used by the Legislative Consultancy of the Federal Chamber to distribute tasks. This system not only categorizes legislative texts thematically but is also being developed to perform automatic translations, identify parliamentarians in images and videos, interpret emotions in Portuguese texts, and answer citizens' queries. Future projects include voice recognition, audio-to-text conversion, keyword extraction, and the synthesis of summaries of legislative documents, as announced by the Press Office of the Chamber of Deputies in 2019.

It is evident that the application of generative AIs to automate steps in the legislative process could be enhanced with parallel mechanisms for systematization, sharing, and intensive use of this knowledge produced and required by parliamentary activity.

It is emphasized that the potential utility of AI in various sectors of the Legislative Branch, especially to improve the work of legislative houses, including parliamentary advisory and legislative consultancy. AI could be used to organize meetings, control deadlines, and monitor the compliance with regimental and constitutional rules, increasing the efficiency of legislative procedures (Agostini, 2020).

Artificial intelligence (AI) plays a crucial role in lawmaking, potentially improving legislative quality and reducing legal uncertainty by defining criteria for creating algorithms that follow the guidelines established by Complementary Law n°. 95 of 1998. This law guides the elaboration, writing, and consolidation of laws, aiming for clearer and more precise texts. Additionally, legislative technique, enriched by the collection and analysis of real data by AI, can standardize and enhance the formulation of legislative propositions. Understanding the formal structure of laws, which goes beyond visual organization to ensure clarity and precision, is essential for the comprehension and improvement of legal texts.

Moreover, it's important to understand the mandatory nature of complying with the rules imposed by the internal regulations of the Legislative Houses, as part of the legislative procedure aimed at safeguarding the production of provisions to achieve full conditions of regularity. For this, it is fundamentally necessary to observe constitutional and regimental law with the settlement on the possibility of triggering the judicial control of constitutionality and regularities of the legislative process and the laws produced without the observance of due legislative process (Silva, 2007).

Finally, one must not forget that the legislative process must be covered by the effectiveness of the preventive control of constitutionality (Del Negri, A. L. 2003).

However, an examination of the legislative procedure that legitimizes the exercise of law production must be in accordance with the constitutional process, shaping the due legislative process (Silva, 2007).

Thus, using generative AI would facilitate the identification of flaws in the regimental rules of the Legislative Assemblies or City Councils related to the lack of contradiction, ample defense, and equality.

Additionally, the importance of Artificial Intelligence (AI) in enriching interdisciplinary research is emphasized, fundamental to supporting technical opinions in permanent commissions and enriching debates in parliament. AI is also seen as a means of enhancing participatory democracy, making the legislative process more engaging and transparent, as discussed by Agostini in 2020.

On the other hand, Agostini stresses the significant influence that analyzing and reacting to parliamentary statements and actions on various debated topics can have on democratic representativeness. This influence is maximized when supported by technical precision and confidentiality, guaranteed by the use of advanced technological intelligence tools. This contributes to enhancing debates in legislative houses, promoting greater efficacy and depth in discussions.

The integration of digital communication and Artificial Intelligence (AI) in legislative activities, as highlighted by Lucena in 2023, not only revolutionizes parliamentary marketing but also improves the quality and efficiency in content production and monitoring of parliamentary routines. This technology allows for the creation of innovative projects and the optimization of tasks, transforming routine activities into notably impactful and effective processes.

Araújo et al. (2020) highlight the need to align the adoption of artificial intelligence (AI) in public administration, including the Legislative Branch, with democratic principles and the collective interest. They emphasize the importance of establishing ethical and legal guidelines for the responsible use of these technologies, paying attention to risks such as algorithmic biases and misinformation.

According to the authors, legislative modernization through AI should aim for public benefits and consider democratic implications, ensuring transparency and focusing on the common good (Araújo et al. 2020).

It is interesting to note that, in general, the main requirements of AI ethics are related to aspects such as reflection on the objectives and purposes of research, the direction of research funding, the connection between science and politics, the security of AI systems, the responsibility links underlying the development and use of AI technologies, the inscription of values in technical artifacts, the

orientation of the technology sector towards the common good, and much more (Daly et al., 2019).

3 Methodological procedures

The integrative literature review (Botelho, Cunha, & Macedo, 2011; Whittemore & Knafl, 2005; Tranfield, Denyer & Smart, 2003) was the methodology used for this scientific work. It is important to highlight that the literature review is a first step towards the construction of scientific knowledge, as it is through this process that new theories emerge, as well as gaps and opportunities for research on a specific subject are recognized (Pereira & Cunha, 2020).

Thus, the literature review process requires the elaboration of a synthesis based on different topics, capable of creating a broad understanding of the knowledge. Furthermore, it is necessary to emphasize that integrative literature reviews are the broadest type of review, as they allow for the simultaneous inclusion of experimental and non-experimental research, in order to more fully understand the phenomenon to be investigated (Whittemore & Knafl, 2005; Pereira & Cunha, 2020).

Tranfield, et al. (2003) propose three steps for the integrative literature review: planning the review; conducting the review; and reporting and disseminating the review results.

STEP 1 - PLANNING THE REVIEW: The research began with the identification of the need for the review and contextualization of the theme. The definition and contextualization of the theme delimit and specify its content, avoiding ambiguity that could lead to a deviation from the research focus (Pereira & Cunha, 2020).

In the last phase of step 1, the review protocol was constructed. The protocol is a plan that contributes to the objectivity of the research, through an explicit description of the steps that were followed.

STEP 2: CONDUCTING THE REVIEW: In this step, the following were defined: search strategy; databases; time period; languages of the articles; terms to be searched; tools for collecting and organizing information; and the inclusion and exclusion criteria.

The search string ("artificial intelligence" OR "generative artificial intelligence" OR "AI" OR "GAI") AND ("legislative branch" OR "legislative power" OR "legislative assembly" OR "parliament" OR "congress") AND ("applications" OR "uses" OR "utility" OR "implementation") underpinned the searches in the Scopus database, which was selected for offering the broadest documentary coverage. However, to increase the level of reliability, the results were also cross-referenced with the Web of Science Core Collection and Ebsco to confirm the inclusion of all relevant

studies. The search was conducted by title, abstract, and keyword, and located a total of 170 results.

In the databases, the following criteria for inclusion and exclusion of articles were applied: articles produced in the last 5 years (2018 to 2023), language: English and Spanish, and types of documents: articles, totaling a pre-selection of 170 articles.

In the next step, the title and abstract were read to identify adherence to the research theme and remove duplicated articles, thus selecting 21 articles for full reading. After reading, the strategy of a synthesis matrix was used for data extraction, used for analysis and discussion of the results. The findings are inserted in Table 2.

No discrepancies were detected, confirming Scopus and Ebsco as the most comprehensive data sets. After several iterations and based on the research questions of the present study, the following research sequence was identified as the broadest and most effective for retrieving relevant articles.

STEP 3: REPORTING AND DISSEMINATION: The report was prepared from the analysis and discussion of the results as follows.

Table 1 - Flowchart of the Research Phases(Integrative Literature Review)

Stage 1 Revision Planning	Stage 2 Conducting the Review	Stage 3 Reporting - Dissemination
Identification of the need for revision and contextualization of the theme	Definition of the search strategy, databases, time period, languages of the articles, terms to be searched, tools for collecting and organizing information, and inclusion and exclusion criteria.	Definition of the search strategy, databases, time period, languages of the articles, terms to be searched, tools for collecting and organizing information, and inclusion and exclusion criteria.
Construction of the review protocol.	Use of the search string and selection of databases (Scopus, Web of Science Core Collection, Ebsco).	
	Application of inclusion and exclusion criteria (articles from the last 5 years in English and Spanish).	
	Reading titles and abstracts, removing duplicates, and selecting articles for complete reading.	
	Use of a synthesis matrix for analysis and discussion of results.	

Source: survey data (2023)

4 Analysis and discussion of results

This section presents the results obtained from the analysis and synthesis of the 21 articles listed in Table 2. The purpose of the synthesis was to identify themes that could answer the question regarding the use of generative artificial intelligences in the Legislative Power. Data synthesis was operationalized with the aid of thematic analysis proposed by Braun and Clarke (2016). The thematic analysis allowed the themes to be identified and coded throughout the entire process of reading and analyzing the articles.

Table 2. Summary of the results of the integrative literature review

Author	The use of generative AIs in the Legislative Branch
(Fischer & Piskorz-Ryń, 2021)	The guidelines said that in order to "obtain trustworthy artificial intelligence" the following three elements were required: it should be (i) in compliance with the rules of the law; (ii) be developed with respect to ethical standards; and (iii) be trustworthy.
(Ahmed et al., 2023)	Automated Knowledge Organization in relation to the AI/ML-based Subject Indexing System for Libraries
(Nowik, 2021)	Algorithm, like algorithmic management, is not only a tool for making more effective decisions, but also an ethical and legal category. This is because it is the result of people's knowledge and will, even if their actions are not entirely predictable. The algorithm can therefore lead to wrong decisions, discrimination, and even social exclusion (algorithmic exclusion).
(Fitsilis, 2021)	In April 2021, the Committee for the Future of the Parliament of Finland (Eduskunta) organised an extraordinary hearing, i.e. of an artificial intelligence (AI). While some legislatures and research groups had already begun to study the implication of AI in the parliamentary realm, this took the parliamentary world by surprise. It was the first time a parliament had directly interacted with an AI system in a real parliamentary process. This research note attempts to conduct a preliminary analysis on the experiment and discusses its implications for future actions in the development of parliamentary tools and services using AI-based technologies. The analysis is dedicated to the intra-parliamentary workspace, considering possible effects on the main functions of the parliament, such as legislating and oversight. The note aims to stimulate the discussion around the implementation strategy, but also for the regulation of such systems in the parliamentary environment.
(Stuurman & Lachaud, 2022)	AI regulation is emerging in the EU. European authorities, NGOs and academics have already issued a series of proposals to accommodate the "development and adoption of AI" with an "appropriate ethical and legal framework" and promote what the European Commission has called an "ecosystem of trust."

(Silveira, 2020).	It analyzes the discourse on algorithmic machine learning and deep learning systems, especially on artificial neural networks, presented as inscrutable, unfathomable and incomprehensible. It also exposes the criticism and possible ineffectiveness of transparency to support the explainability of artificial intelligence solutions.
(A Comissão para o Futuro ouviu a AI provavelmente como a primeira comissão parlamentar do mundo, [s.d.])	The Committee for the Future analysed the importance of technologies such as artificial intelligence as both an opportunity and a threat. The goal was not to find the best possible answers to problems, but to provide examples that are as illustrative as possible of how artificial intelligence deals with problematic questions and how it responds to them.
(Salgado-Criado, J., & Fernández-Aller, C., 2021)	This article was written before the European Commission published its proposal for a regulation on artificial intelligence (AI). In a first and provisional analysis of the proposed regulation, we note that the proposed regulation incorporates some of the basic principles set out in our article: it prioritizes fundamental rights and incorporates some human rights principles, such as accountability and the inclusion of governance through supervisory authorities to implement and enforce the regulation
(Pilving, I, 2023)	Despite the image of developed e-governance, advanced automated decision-making (ADM) systems have not been widely used in Estonian public administration and there is still no general legal framework for them.
(Martin, 2022)	An approximation of some EU initiatives on the regulation of liability for damages caused by AI systems.
Artificial Intelligence: Innovation in parliaments. ([s.d.]).	The U.S. House of Representatives has introduced an AI tool to automate the process of analyzing differences between bills, amendments, and current laws, supporting legislative staff to more easily see the impacts of amending provisions contained in bills they move through the legislative process. Using NLP technology, the tool is able to interpret citations to legal provisions described in bills, and retrieve them and interpret and execute modifying instructions. Already available to the Legislative Advisory, the tool currently operates at an accuracy level of 90, with human feedback improving this the more the system is used. The National Graduate Institute for Policy Studies (GRIPS) in Tokyo has developed a system using data from the Diet of Japan to make parliamentary videos searchable, using the latest speech recognition techniques to match parliamentary video feeds and official minutes. Based on this system, GRIPS is currently developing an integrated speech recognition and audio and visual information analysis system. The introduction of machine learning and AI in the analysis of contextual and visual information is applied in this process to develop an interface of text, audio and video records, allowing the analysis, aggregation and visualization of parliamentary digital communications, which eventually allows the automatic selection of (relevant) highlights of parliamentary debates.

The OECD Global Parliamentary group on AI - OECD.AI, [s.d.]	The Network is a learning hub for legislators and parliamentary staff and works in partnership with parliamentary assemblies, organisations and the European Parliament. Its flexible structure allows legislators from across the policy spectrum to share experiences, identify best practices and promote international legislative cooperation. In October 2019, Network members called for the creation of a thematic parliamentary group on Artificial Intelligence, with a network and educational component (technical and political) to help inform national legislative processes.
D'Acquisto, G. (Ano).	Governing the autonomy of machines may be a very ambitious goal for humans, since machines are geared first to the principles of formal logic and then – possibly – to ethical or legal principles.
(Von Lucke et al., 2023)	Parliamentary artificial intelligence (AI) and AI-based applications are beginning to find their way into the institutional workspace. In the parliamentary routine, AI-based application systems would, for example, be able to automatically become aware of certain events, notify third parties, recommend actions, make predictions, initiate precautionary measures and make certain decisions even without the involvement of their users, almost in real time.
Responsabilidade por danos causados pelo uso de tecnologias de inteligência artificial, [s.d.]	Artificial intelligence technologies, which have recently been developing rapidly, along with indisputable advantages, also create many dangers, the implementation of which causes harm. The reparation of such damage raises questions concerning the subjects, the act itself that caused the damage, causality, etc. The situation is also complicated by the imperfection of the legal regulation of relationships on the use of artificial intelligence technologies and the insufficiency or ambiguity of judicial practice on the reparation of damages caused by means of digital technologies.
(Lischka, 2019)	Digital platforms are under public scrutiny and face a negotiation process regarding their social responsibility. In this institutional discourse on moral responsibilities, corporate actors seek to defend the legitimacy of digital platforms. This study conceptualizes strategic communication as a means of conveying conformity to expectations of legitimacy, shaping judgments of legitimacy, and discursively conducting institutional work.
(Roksandić, S., Protrka, N., & Engelhart, M. 2022)	From all kinds of industry, communication, education, banking, government, services, manufacturing, medicine, and more, the applications of Artificial Intelligence (hereinafter: AI) can be found in many sectors of our life. Public safety and criminal justice are gaining advantages thanks to artificial intelligence.
(Lucke et al., 2023)	Parliaments around the world are examining the use of applications that use artificial intelligence (AI) technologies to better perform certain tasks. In view of the conceivable tools, areas of application, use scenarios and needs, AI-induced changes in parliaments can be expected in the coming years. It is all the more remarkable that the introduction of AI in parliaments is a topic that is generally under-researched. The analysis provides insight into the prioritization of AI-based technologies in the parliamentary environment. The study examined

	the relevance and priority of more than 210 applications and topics of AI technologies in Parliament for different parliamentary areas, including 36 proposals around legislative powers and procedures, referred to as "legislation". The most important results related to the legislation are presented and discussed.
(Astaiza, W. E. U., 2023)	The advancement of technology with regard to artificial intelligence (AI) has led academia to question not only how far it can go, but also what are the possible harmful effects that it can bring to humanity, which is why, in the absence of legal regulation on the subject, the Presidency of the Republic has published the "Ethical Framework for Artificial Intelligence in Colombia" in response to the global dynamics of AI.
(Battaglia & Di Vetta, 2022)	Human action has always been intertwined with technologies. Given this social aspect, the mastery of life sciences and technology is not a matter of individualistic ethics. On the contrary, it includes the construction of a more integrated political society. For these reasons, a modern analysis of the ethical impact of BCIs should encompass an idea of how to deal with emerging technologies with a view to making room for a stronger civic hand in shaping our future.
(Ronanki, 2023)	Ethical guidelines are an asset for the development of artificial intelligence (AI) and compliance with them will soon be a procedural requirement once the EU AI Act is ratified in the European Parliament. However, developers often lack explicit knowledge on how to apply these guidelines during the system development process. A literature review of different ethical guidelines from various countries and organizations revealed inconsistencies in the principles presented and in the terminology used to describe such principles.

Source: survey data (2023)

As a key point for discussion, it's important to assert that the integration of generative artificial intelligence (AI) in the legislative process represents a frontier of innovation with the potential to significantly transform the efficiency, accessibility, and quality of parliamentary work.

This study revealed how automation and advanced information processing can assist in the management of legislative proposals, the analysis of large data volumes, and the facilitation of public participation. Comparing these findings with the reviewed literature, a global trend towards the adoption of AI technologies in the public sector is observed, albeit with significant variations in terms of application, challenges faced, and results achieved.

In emerging countries, the application of AI in the legislative process demonstrates significant innovative potential, despite being in initial stages compared to developed countries. A notable example is the 'Ulysses' system, implemented by the Chamber of Deputies in Brazil, illustrating how technology can be used to automatically organize legislative documents and facilitate communication with the public. This case reflects the challenges and

opportunities faced by emerging countries in implementing AI technologies, where infrastructure and resource limitations can be compensated by the willingness to innovate and improve legislative processes.

The Brazilian experience highlights the ability of emerging countries to adopt advanced technological solutions to promote efficiency, transparency, and citizen participation in the legislative process, despite obstacles.

In developed countries, the integration of AI in the legislative process often benefits from advanced technological infrastructure and greater availability of resources, allowing for a broader and more sophisticated application of these technologies. These initiatives, while more established, face similar challenges in terms of ethics, privacy, and the need for clear regulation, emphasizing the universal importance of cautious and responsible approaches in the implementation of AIs in the public sector.

This study contributes to the existing literature by highlighting how the application of generative AIs in the legislative process can be adapted to different national contexts, considering the technical, cultural, and political specificities of each country. Moreover, it underscores the importance of integrated approaches that not only implement advanced technologies but also promote policies and practices that ensure their ethical and responsible use, aligned with democratic principles and citizens' rights.

Indeed, the implementation of generative artificial intelligence (AI) in the legislative process raises an important debate about its potential effects on democratizing access to the law-making process and promoting participatory democracy. On one hand, AI technologies offer the promise of making legislative processes more efficient and accessible, enabling greater citizen participation through interactive digital platforms that facilitate the understanding of legislative activities and encourage public engagement.

It must also be considered that concerns emerge about the ethical, social, and political implications of these technologies. Issues such as algorithmic bias, transparency in automated decisions, and the need for human oversight are critical to ensuring that the implementation of AI not only benefits the legislative process in terms of efficiency but also strengthens democratic principles and preserves citizens' rights.

In this sense, the present article seeks to offer a detailed and balanced view on how generative artificial intelligence influences the legislative process and participatory democracy, highlighting the importance of adopting these technologies ethically and responsibly.

Furthermore, as a result of the integrative literature review, some important implications regarding the use of generative artificial intelligences (AIs) in the

Legislative Branch are extracted, among which, in summary, the following stand out:

- Scarcity of studies and practical applications - There is a notable lack of research and practical applications exploring the use of generative AI in the Legislative Power, indicating a significant opportunity for future investigations;
- Revolutionary potential of generative AIs - Generative AIs have the potential to optimize the legislative process, from analyzing large volumes of data to increasing transparency and citizen participation;
- Ethical and technical challenges - The integration of generative AI in the Legislature presents ethical and technical challenges, including issues of privacy, algorithmic bias, and the need for clear regulations;
- Interdisciplinary dialogue - An interdisciplinary dialogue involving legislators, technology experts, ethics, and civil society is fundamental to formulate responsible and ethical strategies;
- Essential legal and ethical compliance - Compliance with existing laws and adherence to ethical principles are crucial for the implementation of AIs in the Legislature, ensuring respect for citizens' rights and reliability in technologies;
- Transparency and explainability - Transparency and the ability to explain decisions made by AIs are vital for public trust, accountability, and bias identification;
- AI regulation in the EU - The European Union leads in AI regulation by proposing a comprehensive set of rules that can serve as a model for other contexts, emphasizing the importance of fundamental rights and ethical AI implementation;
- Automation of legislative processes with AI - The application of generative AIs can significantly improve innovation and efficiency in the legislative process, contributing to a more effective and participative Legislature;
- Balance between technological innovation and governance - A responsible and ethical approach is essential to explore the benefits of generative AIs in the Legislature, ensuring the use of these technologies in a way that promotes the common good and respects democratic principles.

5 Conclusions

Regarding the Legislative Branch, as observed, a variety of challenges are faced. This was demonstrated by the scarcity of findings in the integrative literature review on generative AIs in the Legislative Branch.

However, a significant advancement is already in practice: the automatic categorization of legislative documents. This is possible thanks to the 'Ulysses' software, developed by the Federal Chamber. This system not only autonomously organizes legislative texts but is also enhancing its capabilities in automatic translation. Moreover, 'Ulysses' has the ability to recognize parliamentarians in images and videos and is equipped to respond to citizens' inquiries related to parliamentary issues. These features are already a reality and represent a significant step in the use of technology within the legislative domain.

Indeed, over time, the enhancement of the Legislative Branch will be necessary to increase representative democracy. Some suggestions for the applicability of information technology in the parliament could be how to use it in the planning phase, in the short term, measures that include voice recognition of the speaker, automatic transformation of audio into text, extraction of keywords from legislative documents to improve research and databases, as well as constructing texts by providing only the necessary parameters for legislative proposals.

Perhaps one day we will see the systematization of proposals for popular initiatives, aiming for a strengthening of participatory democracy and the improvement in the drafting of opinions, notes, or technical consultations in the Parliament prepared by AI.

Effectively, we would have a greater and better popular participation in which a possibility of rejuvenating representative democracy is sought, including to value it among vast sectors of the population, especially those who do not have a daily influence over political power and its decisions, making the legislative process more attractive and transparent.

Certainly, AI would allow, in a simple and effective way, an increase in participatory democracy in decisions, reinforcing the legitimacy of the Legislative Branch with public acceptance, consent, and mutual understanding.

It must not be forgotten that the use of AI is no longer a trend, but a reality. To be adapted to changes, all entities of the federation need to be updated and, most importantly, willing to update themselves.

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